	AzureSky	
	Ensuring Validity of Results (Clause 7.7)	Document no: AS001V
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1. Purpose

To define the systematic approach used by the laboratory to monitor the validity of test results and ensure continuing accuracy, reliability, and traceability in compliance with **ISO/IEC 17025:2017, Clause 7.7**.

2. Scope

This procedure applies to all analytical preparation and determinations performed by the laboratory, including **Preparation & Proximate Analysis (Moisture, Ash, Volatile Matter, and Fixed Carbon)** and **Calorific Value (CV)** testing, as well as to all instruments and reference materials associated with these determinations.

3. Responsibilities

Role	Responsibilities
Laboratory Manager / Supervisor	Approves the monitoring programme, reviews trends, and authorizes corrective actions. Ensure adherence to monitoring frequency, maintain QC logs, verify control chart performance, and review duplicates and CRMs.
Laboratory Technician	Perform verification checks, run duplicates and CRMs, record results accurately, and initiate investigation for out-of-control data.
Operations Manager	Reviews all records for completeness and trends, ensures calibration certificates are current, and reports performance in management review.

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4. Definitions

- **CRM:** Certified Reference Material — a material with a certified analyte value and associated uncertainty.
- **Control Chart:** Graphical tool used to monitor performance trends and detect bias or drift.
- **Repeatability Limit:** Maximum allowable difference between duplicate results under the same conditions.
- **Verification:** Confirmation that an instrument performs within defined limits before use.

5. Monitoring Methods

5.1 Instrument Calibration and Verification

1. Instruments that require calibration — **Ash Furnace, Volatile Furnace, Moisture Oven, Drying Oven, Analytical Balances, and Mass Pieces** — are sent to **SANAS-accredited calibration facilities** to ensure traceability to national standards.
2. **Calibration certificates** are reviewed upon receipt and retained in the instrument calibration file.
3. **Verification prior to use:**
    - Analysts perform daily or batch verification checks (e.g., temperature control and recording, balance verification, equipment efficiency) before starting work.
    - Any deviations beyond control limits are reported to the Laboratory Manager/supervisor before testing proceeds.
4. **CV instrument** (Calorimeter) is **calibrated in-house** using a **Certified Benzoic Acid pill** of known calorific value. Calibration results are documented and plotted on a control chart to assess stability.

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5.2 Analytical Quality Control

1. **Duplicate Analyses:**
    - Each sample run includes at least one duplicate test sample to evaluate precision.
    - Duplicates must meet the following **repeatability limits**:

Parameter	Repeatability Limit
Calorific Value (CV)	≤ 0.12 MJ/kg
Ash (≤ 10 %)	≤ 0.2 % abs
Ash (> 10 %)	≤ 2 % of mean
Volatile (≤ 10 %)	≤ 0.3 % abs
Volatile (> 10 %)	≤ 3 % of mean
Inherent Moisture (≤ 5 %)	≤ 0.1 % abs
Inherent Moisture (> 5 %)	≤ 0.15 % abs

    - Results exceeding limits must be re-run, if it still is out of repeatability limit, run a CRM. If problem persists, do a quick verification/maintenance on instrument. Conduct an investigation if there is any drift found on verification/maintenance.
2. **Certified Reference Material (CRM):**
    - A CRM is analysed **after every 10 routine runs** for each parameter.

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- The measured value is compared with the certified value; deviations beyond control limits (reproducibility) are investigated.

3. **Control Charts:**
   - Control charts are maintained for each parameter using CRM and duplicate data.
   - Each new result is plotted and reviewed to detect bias, drift, or loss of precision.
   - Out-of-control points (beyond Reproducibility limits) or systematic trends require immediate suspension of affected runs until resolution.

5.3 Method Validation and Ongoing Verification

1. All methods are **validated before routine use** to confirm accuracy, precision, linearity, and detection limits.
2. Periodic re-validation is performed after any significant change (New equipment).
3. Analysts verify method performance using QC data trends and inter-analyst comparisons.

5.4 Proficiency Testing and Interlaboratory Comparison

1. The laboratory participates in **proficiency testing (PT)** for each accredited parameter.
2. PT performance is evaluated using **z-scores** and included in management review.
3. If no PT is available, **inter-laboratory comparisons** are conducted

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5.5 Trending and Review

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1. All QC data (duplicates, CRMs, calibrations, PT results) are trended per control chart.
2. The **Laboratory Manager** and **Laboratory Supervisor** jointly review control charts.
3. Any observed bias, drift, or degradation of performance triggers a formal **Investigation**.
4. Persistent trends may result in method re-validation, recalibration, or staff retraining.

6. Corrective Action Procedure

- Stop analysis of samples for affected parameter or batch.
- Re-verify instrument verification, calibrate where necessary and perform impact analysis on affected batches.
- Document root cause, corrective and preventive action in the **Nonconformance Register**.
- Confirm restored control with CRM or duplicate results before resuming routine testing.

7. Records and Documentation

The following records are retained electronically or in bound QC logbooks:

Record	Retention Period	Responsibility
Calibration Certificates	5 years	Operations Manager
Daily Verification Logs	3 years	Analyst
CRM Results & Control Charts	5 years	Supervisor

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Record	Retention Period	Responsibility
Duplicate Test Records	3 years	Analyst
PT/Inter-lab Results	5 years	Operations Manager
IRQs and Corrective Actions	5 years	Quality Manager

8. Review

This procedure shall be reviewed annually during management review meetings.

9. References

- ISO/IEC 17025:2017 — Clauses 7.6, 7.7, 7.8
- ILAC G8:09/2019 — Guidelines on decision rules and reporting
- ASTM D3173, D3174, D3175, D5865 — Coal analysis standards
- Laboratory QMS Manual, Section 7 — Quality Control and Validation

10. Revision

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